

```
import pandas as pd
import numpy as np
```

```
df_energy = pd.read_csv('/content/energy_consumption.csv')
print("Energy consumption data loaded successfully.")
```

Energy consumption data loaded successfully.

```
import pandas as pd

# 1. Create a new datetime column by combining 'year' and 'discrete_month'
df_energy['Date'] = pd.to_datetime(df_energy['year'].astype(int).astype(str) + df_energy['discrete_month'].astype(int))
print("Datetime column 'Date' created.")

# 2. Set this newly created datetime column as the index of the DataFrame
df_energy.set_index('Date', inplace=True)
df_energy.index.name = 'Date'
print("Datetime column set as index and named 'Date'.")

# 3. Sort the df_energy DataFrame by its new datetime index
df_energy.sort_index(inplace=True)
print("DataFrame sorted by datetime index.")

# 4. Check for any duplicate timestamps in the index and handle if necessary
# Using groupby(level=0).mean() to aggregate all duplicate rows by their month
# which will result in a DataFrame with a unique index.
if df_energy.index.duplicated().any():
    print(f"Warning: {df_energy.index.duplicated().sum()} duplicate timestamps found.")
    df_energy = df_energy.groupby(level=0).mean()
    df_energy.sort_index(inplace=True) # Sort again after aggregation
    print("Duplicate timestamps handled.")
else:
    print("No duplicate timestamps found.")

# 5. Resample the data to ensure a consistent monthly frequency
# Create a complete date range to reindex against
full_date_range = pd.date_range(start=df_energy.index.min(),
                                end=df_energy.index.max(),
                                freq='MS') # 'MS' for Month Start frequency

# Reindex the DataFrame to fill any missing months. For numerical columns,
df_energy_resampled = df_energy.reindex(full_date_range)

# Check if resampling introduced missing values in key columns like 'Electricity_consumption'
missing_after_resample = df_energy_resampled['Electricity_consumption'].isna().sum()
if missing_after_resample > 0:
```

```
print("After resampling, {missing_after_resample} missing values found")
print("These indicate missing months in the original data range. These")
else:
    print("No missing months detected or introduced by resampling within the")

df_energy = df_energy_resampled.copy()
print("DataFrame resampled to ensure consistent monthly frequency.")

print("\nFirst 5 rows of preprocessed df_energy:")
display(df_energy.head())
print("\nLast 5 rows of preprocessed df_energy:")
display(df_energy.tail())
print("\nInfo of preprocessed df_energy:")
df_energy.info()
```

```
import matplotlib.pyplot as plt
from statsmodels.tsa.seasonal import seasonal_decompose
from statsmodels.tsa.stattools import adfuller

# Ensure the 'Electricity_consumption' column is available in df_energy
if 'Electricity_consumption' not in df_energy.columns:
    raise KeyError("'Electricity_consumption' column not found in df_energy")

# 1. Plot the 'Electricity_consumption' column
plt.figure(figsize=(12, 6))
plt.plot(df_energy.index, df_energy['Electricity_consumption'])
plt.title('Electricity Consumption Over Time')
plt.xlabel('Date')
```

```

plt.ylabel('Electricity Consumption')
plt.grid(True)
plt.show()

# 2. Decompose the time series (given monthly data, period=12 for annual seasonality)
# Handle potential missing values in the series for decomposition
series_for_decomposition = df_energy['Electricity_consumption'].dropna()

# Ensure enough data points for decomposition with period=12
# seasonal_decompose requires at least 2*period data points for 'additive' model
# and 3*period for 'multiplicative' model. For period=12, need at least 24 data points
if len(series_for_decomposition) < 2 * 12: # Check if there are at least 24 data points
    print(f"\nWarning: Not enough data points ({len(series_for_decomposition)} data points).")
    print("Skipping seasonal decomposition.")
else:
    print("\nPerforming seasonal decomposition...")
    try:
        decomposition = seasonal_decompose(series_for_decomposition, model='additive')
        fig = decomposition.plot()
        fig.set_size_inches(10, 8)
        plt.suptitle('Time Series Decomposition of Electricity Consumption')
        plt.tight_layout(rect=[0, 0.03, 1, 0.98])
        plt.show()
        print("Seasonal decomposition completed.")
    except Exception as e:
        print(f"Error during seasonal decomposition: {e}")
        print("This might happen if there are too few non-null data points")

# 3. Perform Augmented Dickey-Fuller test for stationarity
print("\nPerforming Augmented Dickey-Fuller test for stationarity...")
# The test requires no NaNs, so use dropna()
adf_test_result = adfuller(df_energy['Electricity_consumption'].dropna())

print(f'ADF Statistic: {adf_test_result[0]}')
print(f'p-value: {adf_test_result[1]}')
print('Critical Values:')
for key, value in adf_test_result[4].items():
    print(f'\t{key}: {value}')

if adf_test_result[1] <= 0.05:
    print("Conclusion: The series is likely stationary (reject H0).")
else:
    print("Conclusion: The series is likely non-stationary (fail to reject H0).")

Info of preprocessed df_energy:
<class 'pandas.core.frame.DataFrame'>
DatetimeIndex: 32 entries, 2017-05-01 to 2019-12-01
Freq: MS
Data columns (total 22 columns):
#   Column                                     Non-Null Count  Dtype
---  -

```

```

0 Electricity_consumption      32 non-null    float64
1 Normalized_consumption      32 non-null    float64
2 Community_ID                 32 non-null    float64
3 Temperature                  32 non-null    float64
4 NumOfConnected-COMMELEC      32 non-null    float64
5 NumOfConnected-FREENRESIE    32 non-null    float64
6 NumOfConnected-GOVTELEC      32 non-null    float64
7 NumOfConnected-INDTELEC      32 non-null    float64
8 NumOfConnected-RESIEXPE      32 non-null    float64
9 NumOfConnected-TOTAL         32 non-null    float64
10 Expat_Ratio                 32 non-null    float64
11 Multi.Storey.Building        32 non-null    float64
12 Multi.Storey.Ratio.Building  32 non-null    float64
13 Investment.Villa            32 non-null    float64
14 Public.Building              32 non-null    float64
15 Industrial.Building          32 non-null    float64
16 Total.Buildings              32 non-null    float64
17 population                   32 non-null    float64
18 building_occupancy           32 non-null    float64
19 continuous_month             32 non-null    float64
20 discrete_month              32 non-null    float64
21 year                        32 non-null    float64
dtypes: float64(22)
memory usage: 5.8 KB

```

Reasoning: The next step is to determine the optimal SARIMA parameters using `pmdarima.auto_arima`. This function is not part of standard Anaconda or Colab

installations and needs to be installed first. It is best practice to install external libraries in a separate code block.

```
import sys

# Check if pmdarima is installed, and install if not
try:
    import pmdarima as pm
    print("pmdarima is already installed.")
except ImportError:
    print("pmdarima not found. Installing pmdarima...")
    !{sys.executable} -m pip install pmdarima
    import pmdarima as pm
    print("pmdarima installed successfully.")
```

pmdarima not found. Installing pmdarima
Collecting pmdarima
Downloading pmdarima-2.1.1-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.whl (689.1 kB)
Requirement already satisfied: joblib>=0.11 in /usr/local/lib/python3.12/dist-packages (0.22.0)
Requirement already satisfied: cython!=0.29.18, >=0.29 in /usr/local/lib/python3.12/dist-packages (0.29.35)
Requirement already satisfied: numpy>=1.21.6 in /usr/local/lib/python3.12/dist-packages (1.26.4)
Requirement already satisfied: pandas>=0.19 in /usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: scikit-learn>=0.22 in /usr/local/lib/python3.12/dist-packages (1.5.2)
Requirement already satisfied: scipy>=1.13.0 in /usr/local/lib/python3.12/dist-packages (1.13.1)
Requirement already satisfied: statsmodels>=0.14.5 in /usr/local/lib/python3.12/dist-packages (0.14.5)
Requirement already satisfied: urllib3 in /usr/local/lib/python3.12/dist-packages (2.2.3)
Requirement already satisfied: setuptools!=50.0.0, >=42 in /usr/local/lib/python3.12/dist-packages (75.8.2)
Requirement already satisfied: packaging>=17.1 in /usr/local/lib/python3.12/dist-packages (25.0)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (2025.1)
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (3.5.0)
Requirement already satisfied: patsy>=0.5.6 in /usr/local/lib/python3.12/dist-packages (0.5.6)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (1.17.0)
Installing collected packages: pmdarima
Successfully installed pmdarima-2.1.1
pmdarima installed successfully.

Reasoning: With `pmdarima` installed, the next logical step is to use its `auto_arima` function to automatically determine the optimal SARIMA parameters (p, d, q, P, D, Q, S). This fulfills the second instruction of the subtask.

```
import pmdarima as pm

# 2. Determine SARIMA Parameters (p, d, q, P, D, Q, S) using auto_arima
print("Determining optimal SARIMA parameters using auto_arima...")

# The series_for_decomposition was already created and dropped NaNs in the prev
```

```

# Use this series for auto_arima to ensure consistency and avoid issues with Na
# Ensure m=12 for monthly seasonality.
# We will set a relatively low max_order for both p,q,P,Q given the short time
# Also, d and D are set to 0 as the ADF test indicated stationarity.
# However, auto_arima can also search for d and D, so we can let it determine.

# Given the short length of the series (32 data points), setting d and D limits
# or max_d/max_D to 1 is a reasonable approach to avoid overfitting.
# max_p, max_q, max_P, max_Q are also kept relatively low.

# The 'suppress_warnings' argument is added to avoid excessive warnings during
# Using n_jobs=-1 to speed up the process by using all available CPU cores.

# Ensure the series has a DatetimeIndex for auto_arima
if not isinstance(series_for_decomposition.index, pd.DatetimeIndex):
    series_for_decomposition.index = pd.to_datetime(series_for_decomposition.ir

# Fit auto_arima model
# Max order for p, q, P, Q are set conservatively due to short series length.
# D (seasonal differencing) can be 0 or 1.
# d (non-seasonal differencing) can be 0 or 1.
# We set 'start_p', 'start_q', 'start_P', 'start_Q' to 0 to explore all options
# We include 'error_action="ignore"' to skip models that fail to converge, 'ste

# For short series like 32 data points, it's safer to specify max_d, max_D expl
# Since ADF test shows stationary, d=0 might be appropriate, but auto_arima car

# Note: auto_arima on a very short series can be unreliable or might pick a cor
# We should keep this in mind for evaluation.

optimal_model = pm.auto_arima(series_for_decomposition,
                              start_p=0, start_q=0,
                              test='adf',          # use adftest to find optimal 'd'
                              max_p=3, max_q=3,    # maximum p and q
                              m=12,                # frequency of series
                              start_P=0, start_Q=0,
                              max_P=2, max_Q=2,    # maximum P and Q
                              D=1,                  # seasonal differencing
                              d=None,               # let auto_arima determine d, or
                              trace=True,
                              error_action='ignore',
                              suppress_warnings=True,
                              stepwise=True,
                              n_jobs=-1)           # use all available cores

print("\nOptimal SARIMA model parameters:")
print(optimal_model.summary())

# Extracting the optimal parameters
optimal_order = optimal_model.order
optimal_seasonal_order = optimal_model.seasonal_order

```

```
print(f"\nNon-seasonal order (p,d,q): {optimal_order}")
print(f"Seasonal order (P,D,Q,S): {optimal_seasonal_order}")
```

Determining optimal SARIMA parameters using auto_arma...

Performing stepwise search to minimize aic

```
ARIMA(0,1,0)(0,1,0)[12] : AIC=615.244, Time=0.09 sec
/usr/local/lib/python3.12/dist-packages/pmdarima/arma/_validation.py:76: UserWarning:
  warnings.warn('stepwise model cannot be fit in parallel (n_jobs=%i). '
ARIMA(1,1,0)(1,1,0)[12] : AIC=611.353, Time=0.25 sec
ARIMA(0,1,1)(0,1,1)[12] : AIC=607.927, Time=0.37 sec
ARIMA(0,1,1)(0,1,0)[12] : AIC=613.444, Time=0.11 sec
ARIMA(0,1,1)(1,1,1)[12] : AIC=609.756, Time=0.56 sec
ARIMA(0,1,1)(0,1,2)[12] : AIC=609.736, Time=0.49 sec
ARIMA(0,1,1)(1,1,0)[12] : AIC=609.817, Time=0.64 sec
ARIMA(0,1,1)(1,1,2)[12] : AIC=611.663, Time=3.55 sec
ARIMA(0,1,0)(0,1,1)[12] : AIC=609.998, Time=0.25 sec
ARIMA(1,1,1)(0,1,1)[12] : AIC=609.770, Time=0.62 sec
ARIMA(0,1,2)(0,1,1)[12] : AIC=inf, Time=1.25 sec
ARIMA(1,1,0)(0,1,1)[12] : AIC=609.943, Time=0.29 sec
ARIMA(1,1,2)(0,1,1)[12] : AIC=inf, Time=0.72 sec
ARIMA(0,1,1)(0,1,1)[12] intercept : AIC=609.294, Time=0.46 sec
```

Best model: ARIMA(0,1,1)(0,1,1)[12]

Total fit time: 9.716 seconds

Optimal SARIMA model parameters:

SARIMAX Results

```
=====
Dep. Variable: y No. Observations:
Model: SARIMAX(0, 1, 1)x(0, 1, 1, 12) Log Likelihood
Date: Fri, 06 Feb 2026 AIC
Time: 08:14:37 BIC
Sample: 05-01-2017 HQIC
        - 12-01-2019
Covariance Type: opg
=====
              coef      std err          z      P>|z|      [0.025      0.975]
-----
ma.L1          -0.4026        0.324      -1.242      0.214      -1.038        0.233
ma.S.L12        -0.5919        0.532      -1.113      0.266      -1.634        0.450
sigma2          3.225e+12      2.02e-13      1.6e+25      0.000      3.23e+12      3.23e+12
=====
Ljung-Box (L1) (Q):          1.02  Jarque-Bera (JB):          4
Prob(Q):              0.31  Prob(JB):          0
Heteroskedasticity (H):      0.74  Skew:          -1
Prob(H) (two-sided):      0.73  Kurtosis:          3
=====
```

Warnings:

```
[1] Covariance matrix calculated using the outer product of gradients (complex-s
[2] Covariance matrix is singular or near-singular, with condition number 2.09e+
```

Non-seasonal order (p,d,q): (0, 1, 1)

Seasonal order (P,D,Q,S): (0, 1, 1, 12)

Reasoning: The next instruction is to split the data into training and testing sets. I will create a code block to divide the `series_for_decomposition` into training and testing sets, using the last 6 months for testing as per the instruction.

```
import pandas as pd

# 3. Split Data into Training and Testing Sets
# Given the limited number of data points (32 months),
# use the last 6 months for the testing set.

test_size = 6 # Using the last 6 months for testing

train_data = series_for_decomposition[:-test_size]
test_data = series_for_decomposition[-test_size:]

print(f"Training data length: {len(train_data)}")
print(f"Testing data length: {len(test_data)}")

print("\nLast 5 training data points:")
print(train_data.tail())
print("\nFirst 5 testing data points:")
print(test_data.head())
```

```
Training data length: 26
Testing data length: 6

Last 5 training data points:
2019-02-01    5.172288e+06
2019-03-01    6.267848e+06
2019-04-01    8.381489e+06
2019-05-01    8.982221e+06
2019-06-01    1.150411e+07
Freq: MS, Name: Electricity_consumption, dtype: float64

First 5 testing data points:
2019-07-01    1.272007e+07
2019-08-01    1.212832e+07
2019-09-01    1.183052e+07
2019-10-01    1.028516e+07
2019-11-01    4.998199e+06
Freq: MS, Name: Electricity_consumption, dtype: float64
```

Reasoning: The next step is to train the SARIMA model using the optimal parameters (p, d, q, P, D, Q, S) identified by `auto_arima` on the `train_data`, which fulfills the fourth instruction of the subtask.

```

from statsmodels.tsa.statespace.sarimax import SARIMAX

# 4. Train the SARIMA Model
print("Training the SARIMA model...")

# Retrieve the optimal parameters found by auto_arima
# optimal_order = (p, d, q)
# optimal_seasonal_order = (P, D, Q, S)

# Instantiate the SARIMAX model
# The 'enforce_stationarity=False' and 'enforce_invertibility=False' arguments
# used to allow more flexibility in model fitting, especially when auto_arima
# has already determined differencing terms 'd' and 'D'.

model = SARIMAX(train_data,
                 order=optimal_order,
                 seasonal_order=optimal_seasonal_order,
                 enforce_stationarity=False,
                 enforce_invertibility=False)

# Fit the model to the training data
# suppress_warnings is used to clean up the output.
model_fit = model.fit(dispatch=False, low_memory=True)

print("SARIMA model training completed.")
print(model_fit.summary())

```

Training the SARIMA model...
SARIMA model training completed.

SARIMAX Results

```

=====
Dep. Variable:          Electricity_consumption    No. Observations:
Model:                SARIMAX(0, 1, 1)x(0, 1, 1, 12)    Log Likelihood
Date:                  Fri, 06 Feb 2026              AIC
Time:                  08:14:58                      BIC
Sample:                05-01-2017                    HQIC
                     - 06-01-2019
Covariance Type:      approx
=====

```

	coef	std err	z	P> z	[0.025	0.975]
ma.L1	-0.7723	-0	inf	0.000	-0.772	-0.772
ma.S.L12	0	-0	nan	nan	0	0
sigma2	2.637e+12	-0	-inf	0.000	2.64e+12	2.64e+12

```

=====
Ljung-Box (L1) (Q):          nan    Jarque-Bera (JB):
Prob(Q):                    nan    Prob(JB):
Heteroskedasticity (H):      nan    Skew:
Prob(H) (two-sided):        nan    Kurtosis:
=====

```

Warnings:

```
[1] Covariance matrix calculated using numerical (complex-step) differentiation.
[2] Covariance matrix is singular or near-singular, with condition number      inf
/usr/local/lib/python3.12/dist-packages/statsmodels/tsa/statespace/sarimax.py:86:
  warn('Too few observations to estimate starting parameters%s.'
```

```
/usr/local/lib/python3.12/dist-packages/statsmodels/tools/eval_measures.py:414:
  return -2.0 * llf + np.log(nobs) * df_modelwc
/usr/local/lib/python3.12/dist-packages/statsmodels/tools/eval_measures.py:439:
  return -2.0 * llf + 2 * np.log(np.log(nobs)) * df_modelwc
/usr/local/lib/python3.12/dist-packages/statsmodels/tsa/statespace/mlemodel.py:3
  return self.params / self.bse
/usr/local/lib/python3.12/dist-packages/statsmodels/tsa/statespace/mlemodel.py:3
  return self.params / self.bse
/usr/local/lib/python3.12/dist-packages/statsmodels/tsa/stattools.py:472: Runtime
  xo = x - x.mean()
/usr/local/lib/python3.12/dist-packages/numpy/_core/_methods.py:138: RuntimeWarn
  ret = ret.dtype.type(ret / rcount)
```

Reasoning: The SARIMA model training failed with warnings about "Too few observations to estimate starting parameters for seasonal ARMA" and unstable standard errors. This is likely due to the small size of the `train_data` (26 observations) in combination with the seasonal differencing ($D=1$, $m=12$), which further reduces the effective sample size for parameter estimation. To address this, I will adjust the `test_size` from 6 to 3, as allowed by the instructions ("last 3 to 6 months for the testing set"), to increase the training data size and provide more observations for robust model estimation.

```
import pandas as pd

# 3. Split Data into Training and Testing Sets
# Given the limited number of data points (32 months),
# use the last 3 to 6 months for the testing set.

test_size = 3 # Using the last 3 months for testing to maximize training data

train_data = series_for_decomposition[:-test_size]
test_data = series_for_decomposition[-test_size:]

print(f"Training data length: {len(train_data)}")
print(f"Testing data length: {len(test_data)}")

print("\nLast 5 training data points:")
print(train_data.tail())
print("\nFirst 5 testing data points:")
print(test_data.head())
```

```
Training data length: 29
Testing data length: 3
```

```

Last 5 training data points:
2019-05-01    8.982221e+06
2019-06-01    1.150411e+07
2019-07-01    1.272007e+07
2019-08-01    1.212832e+07
2019-09-01    1.183052e+07
Freq: MS, Name: Electricity_consumption, dtype: float64

First 5 testing data points:
2019-10-01    1.028516e+07
2019-11-01    4.998199e+06
2019-12-01    5.834166e+06
Freq: MS, Name: Electricity_consumption, dtype: float64

```

Reasoning: After adjusting the `test_size` to 3, the `train_data` now has more observations (29 instead of 26). I will re-train the SARIMA model with this larger training set, expecting it to mitigate the 'too few observations' warning encountered previously and allow for more stable parameter estimation, thus fulfilling the fourth instruction of the subtask.

```

from statsmodels.tsa.statespace.sarimax import SARIMAX

# 4. Train the SARIMA Model
print("Retraining the SARIMA model with updated training data...")

# Retrieve the optimal parameters found by auto_arima
# optimal_order = (p, d, q)
# optimal_seasonal_order = (P, D, Q, S)

# Instantiate the SARIMAX model
# The 'enforce_stationarity=False' and 'enforce_invertibility=False' arguments
# used to allow more flexibility in model fitting, especially when auto_arima
# has already determined differencing terms 'd' and 'D'.

model = SARIMAX(train_data,
                 order=optimal_order,
                 seasonal_order=optimal_seasonal_order,
                 enforce_stationarity=False,
                 enforce_invertibility=False)

# Fit the model to the training data
# suppress_warnings is used to clean up the output.
model_fit = model.fit(dispatch=False, low_memory=True)

print("SARIMA model retraining completed.")
print(model_fit.summary())

Retraining the SARIMA model with updated training data...
/usr/local/lib/python3.12/dist-packages/statsmodels/tsa/statespace/sarimax.py:86
warn('Too few observations to estimate starting parameters%s.'

```

SARIMA model retraining completed.

SARIMAX Results

```
=====
Dep. Variable:          Electricity_consumption    No. Observations:
Model:                SARIMAX(0, 1, 1)x(0, 1, 1, 12)    Log Likelihood
Date:                  Fri, 06 Feb 2026            AIC
Time:                  08:15:43                    BIC
Sample:                05-01-2017                HQIC
                    - 09-01-2019

Covariance Type:                approx
=====
```

	coef	std err	z	P> z	[0.025	0.975]
ma.L1	-0.4520	5.286	-0.086	0.932	-10.812	9.908
ma.S.L12	-0.8243	0.950	-0.868	0.386	-2.686	1.038
sigma2	2.422e+12	nan	nan	nan	nan	nan

```
=====
Ljung-Box (L1) (Q):                2.00    Jarque-Bera (JB):                0
Prob(Q):                          0.16    Prob(JB):                          0
Heteroskedasticity (H):            nan    Skew:                              0
Prob(H) (two-sided):              nan    Kurtosis:                          1
=====
```

Warnings:

```
[1] Covariance matrix calculated using numerical (complex-step) differentiation.
[2] Covariance matrix is singular or near-singular, with condition number 3.99e+
/usr/local/lib/python3.12/dist-packages/statsmodels/tsa/statespace/mlemodel.py:3
    test_statistic, p_value = breakvar_heteroskedasticity_test(
/usr/local/lib/python3.12/dist-packages/statsmodels/tsa/statespace/mlemodel.py:3
    test_statistic, p_value = breakvar_heteroskedasticity_test(
```

Reasoning: The 'Too few observations' warning persists because the `optimal_order` and `optimal_seasonal_order` were determined using `auto_arma` on the full `series_for_decomposition` (32 observations), while the SARIMA model is subsequently trained on the smaller `train_data` (29 observations). To ensure the chosen parameters are appropriate for the actual training data size and to mitigate the warning, I will re-run `auto_arma` directly on the `train_data` to derive parameters optimized specifically for the training set, thereby addressing the problem at its source and fulfilling the second instruction of the subtask more robustly.

```
import pmdarima as pm

# 2. Determine SARIMA Parameters (p, d, q, P, D, Q, S) using auto_arma
print("Determining optimal SARIMA parameters using auto_arma...")

# Use train_data for auto_arma to ensure the parameters are optimized for the
# This helps in mitigating 'too few observations' warnings during model fitting
# Ensure m=12 for monthly seasonality.
```

```

# Given the short length of the training series, setting d and D limits
# or max_d/max_D to 1 is a reasonable approach to avoid overfitting.
# max_p, max_q, max_P, max_Q are also kept relatively low.

# The 'suppress_warnings' argument is added to avoid excessive warnings during
# Using n_jobs=-1 to speed up the process by using all available CPU cores.

# Ensure the series has a DatetimeIndex for auto_arima
if not isinstance(train_data.index, pd.DatetimeIndex):
    train_data.index = pd.to_datetime(train_data.index)

# Fit auto_arima model
# Max order for p, q, P, Q are set conservatively due to short series length.
# D (seasonal differencing) can be 0 or 1.
# d (non-seasonal differencing) can be 0 or 1.
# We set 'start_p', 'start_q', 'start_P', 'start_Q' to 0 to explore all options
# We include 'error_action="ignore"' to skip models that fail to converge, 'ste

# For short series like 29 data points, it's safer to specify max_d, max_D expl
# Since ADF test shows stationary on the full series, d=0 might be appropriate,

# Note: auto_arima on a very short series can be unreliable or might pick a cor
# We should keep this in mind for evaluation.

optimal_model = pm.auto_arima(train_data,
                              start_p=0, start_q=0,
                              test='adf',          # use adftest to find optimal 'd'
                              max_p=3, max_q=3,    # maximum p and q
                              m=12,                # frequency of series
                              start_P=0, start_Q=0,
                              max_P=2, max_Q=2,    # maximum P and Q
                              D=1,                  # seasonal differencing - can be
                              d=None,               # let auto_arima determine d, or
                              trace=True,
                              error_action='ignore',
                              suppress_warnings=True,
                              stepwise=True,
                              n_jobs=-1)           # use all available cores

print("\nOptimal SARIMA model parameters:")
print(optimal_model.summary())

# Extracting the optimal parameters
optimal_order = optimal_model.order
optimal_seasonal_order = optimal_model.seasonal_order

print(f"\nNon-seasonal order (p,d,q): {optimal_order}")
print(f"Seasonal order (P,D,Q,S): {optimal_seasonal_order}")

```

```

Determining optimal SARIMA parameters using auto_arma...
Performing stepwise search to minimize aic
ARIMA(0,1,0)(0,1,0)[12]           : AIC=512.904, Time=0.02 sec
ARIMA(1,1,0)(1,1,0)[12]           : AIC=512.283, Time=0.14 sec
/usr/local/lib/python3.12/dist-packages/pmdarima/arma/_validation.py:76: UserWarning:
  warnings.warn('stepwise model cannot be fit in parallel (n_jobs=%i). ')
ARIMA(0,1,1)(0,1,1)[12]           : AIC=509.265, Time=0.23 sec
ARIMA(0,1,1)(0,1,0)[12]           : AIC=512.173, Time=0.09 sec
ARIMA(0,1,1)(1,1,1)[12]           : AIC=510.505, Time=0.38 sec
ARIMA(0,1,1)(0,1,2)[12]           : AIC=510.218, Time=0.62 sec
ARIMA(0,1,1)(1,1,0)[12]           : AIC=512.126, Time=0.11 sec
ARIMA(0,1,1)(1,1,2)[12]           : AIC=511.711, Time=0.84 sec
ARIMA(0,1,0)(0,1,1)[12]           : AIC=510.085, Time=0.09 sec
ARIMA(1,1,1)(0,1,1)[12]           : AIC=511.044, Time=0.31 sec
ARIMA(0,1,2)(0,1,1)[12]           : AIC=inf, Time=0.38 sec
ARIMA(1,1,0)(0,1,1)[12]           : AIC=510.908, Time=0.08 sec
ARIMA(1,1,2)(0,1,1)[12]           : AIC=inf, Time=0.30 sec
ARIMA(0,1,1)(0,1,1)[12] intercept : AIC=511.310, Time=0.10 sec

```

Best model: ARIMA(0,1,1)(0,1,1)[12]

Total fit time: 3.705 seconds

Optimal SARIMA model parameters:

SARIMAX Results

```

=====
Dep. Variable:          y      No. Observations:
Model:          SARIMAX(0, 1, 1)x(0, 1, 1, 12)  Log Likelihood
Date:              Fri, 06 Feb 2026      AIC
Time:              08:16:16      BIC
Sample:            05-01-2017      HQIC
                  - 09-01-2019
Covariance Type:          opg
=====

```

	coef	std err	z	P> z	[0.025	0.975]
ma.L1	-0.3735	0.343	-1.089	0.276	-1.046	0.299
ma.S.L12	-0.7449	0.367	-2.028	0.043	-1.465	-0.025
sigma2	2.422e+12	2.09e-13	1.16e+25	0.000	2.42e+12	2.42e+12

```

=====
Ljung-Box (L1) (Q):          0.49      Jarque-Bera (JB):          5
Prob(Q):                  0.48      Prob(JB):                  0
Heteroskedasticity (H):      0.50      Skew:                      -1
Prob(H) (two-sided):        0.46      Kurtosis:                  4
=====

```

Warnings:

```

[1] Covariance matrix calculated using the outer product of gradients (complex-s
[2] Covariance matrix is singular or near-singular, with condition number 2.37e+

```

Non-seasonal order (p,d,q): (0, 1, 1)

Seasonal order (P,D,Q,S): (0, 1, 1, 12)

Reasoning: The previous `auto_arma` run (cell_id: 30388182) successfully determined optimal SARIMA parameters (`optimal_order` and `optimal_seasonal_order`) specifically for the `train_data`. Now, I will use these updated parameters to re-train the SARIMA model with `SARIMAX` on the `train_data` to ensure the model is fit with parameters optimized for its size, thereby fulfilling the fourth instruction of the subtask and addressing the 'too few observations' warning.

```
from statsmodels.tsa.statespace.sarimax import SARIMAX

# 4. Train the SARIMA Model
print("Retraining the SARIMA model with updated training data and parameters...")

# Retrieve the optimal parameters found by the most recent auto_arma on train_
# optimal_order = (p, d, q)
# optimal_seasonal_order = (P, D, Q, S)

# Instantiate the SARIMAX model
# The 'enforce_stationarity=False' and 'enforce_invertibility=False' arguments
# used to allow more flexibility in model fitting, especially when auto_arma
# has already determined differencing terms 'd' and 'D'.

model = SARIMAX(train_data,
                 order=optimal_order,
                 seasonal_order=optimal_seasonal_order,
                 enforce_stationarity=False,
                 enforce_invertibility=False)

# Fit the model to the training data
# suppress_warnings is used to clean up the output.
model_fit = model.fit(dispatch=False, low_memory=True)

print("SARIMA model retraining completed.")
print(model_fit.summary())
```

```
Retraining the SARIMA model with updated training data and parameters...
/usr/local/lib/python3.12/dist-packages/statsmodels/tsa/statespace/sarimax.py:86
  warn('Too few observations to estimate starting parameters%s.')
SARIMA model retraining completed.
```

SARIMAX Results

```
=====
Dep. Variable:          Electricity_consumption    No. Observations:
Model:                SARIMAX(0, 1, 1)x(0, 1, 1, 12)    Log Likelihood
Date:                  Fri, 06 Feb 2026              AIC
Time:                  08:16:28                      BIC
Sample:                05-01-2017                    HQIC
                     - 09-01-2019
Covariance Type:                approx
=====
```

	coef	std err	z	P> z	[0.025	0.975]
=====						

```

-----
ma.L1          -0.4520      5.286      -0.086      0.932      -10.812      9.908
ma.S.L12       -0.8243      0.950      -0.868      0.386      -2.686      1.038
sigma2         2.422e+12      nan        nan        nan        nan        nan
=====
Ljung-Box (L1) (Q):                2.00      Jarque-Bera (JB):                0
Prob(Q):                          0.16      Prob(JB):                      0
Heteroskedasticity (H):            nan      Skew:                          0
Prob(H) (two-sided):              nan      Kurtosis:                     1
=====

```

Warnings:

```

[1] Covariance matrix calculated using numerical (complex-step) differentiation.
[2] Covariance matrix is singular or near-singular, with condition number 3.99e+
/usr/local/lib/python3.12/dist-packages/statsmodels/tsa/statespace/mlemodel.py:3
test_statistic, p_value = breakvar_heteroskedasticity_test(
/usr/local/lib/python3.12/dist-packages/statsmodels/tsa/statespace/mlemodel.py:3
test_statistic, p_value = breakvar_heteroskedasticity_test(

```

```

import pmdarima as pm

# 2. Determine SARIMA Parameters (p, d, q, P, D, Q, S) using auto_arma
print("Determining optimal SARIMA parameters using auto_arma with D=0...")

# Use train_data for auto_arma to ensure the parameters are optimized for
# Ensure m=12 for monthly seasonality.

# Given the short length of the training series, setting max_p, max_q, max_

# The 'suppress_warnings' argument is added to avoid excessive warnings du
# Using n_jobs=-1 to speed up the process by using all available CPU cores

# Ensure the series has a DatetimeIndex for auto_arma
if not isinstance(train_data.index, pd.DatetimeIndex):
    train_data.index = pd.to_datetime(train_data.index)

# Fit auto_arma model
# Max order for p, q, P, Q are set conservatively due to short series lengt
# Setting D=0 to explicitly disable seasonal differencing, as automatic det
# and D=1 led to 'too few observations' warnings during SARIMAX fit.

optimal_model = pm.auto_arma(train_data,
                             start_p=0, start_q=0,
                             test='adf',      # use adftest to find optimal
                             max_p=3, max_q=3, # maximum p and q
                             m=12,            # frequency of series
                             start_P=0, start_Q=0,
                             max_P=2, max_Q=2, # maximum P and Q
                             D=0,              # Explicitly set D=0 to avoid
                             d=None,           # Let auto_arma determine no
                             trace=True,
                             error_action='ignore',

```

```

suppress_warnings=True,
stepwise=True,
n_jobs=-1)          # use all available cores

print("\nOptimal SARIMA model parameters:")
print(optimal_model.summary())

# Extracting the optimal parameters
optimal_order = optimal_model.order
optimal_seasonal_order = optimal_model.seasonal_order

print(f"\nNon-seasonal order (p,d,q): {optimal_order}")
print(f"Seasonal order (P,D,Q,S): {optimal_seasonal_order}")

```

```

Determining optimal SARIMA parameters using auto_arima with D=0...
Performing stepwise search to minimize aic
ARIMA(0,0,0)(0,0,0)[12] intercept : AIC=940.399, Time=0.06 sec
/usr/local/lib/python3.12/dist-packages/pmdarima/arma/_validation.py:76: UserWarning:
  warnings.warn('stepwise model cannot be fit in parallel (n_jobs=%i). '
ARIMA(1,0,0)(1,0,0)[12] intercept : AIC=912.857, Time=1.56 sec
ARIMA(0,0,1)(0,0,1)[12] intercept : AIC=925.454, Time=0.31 sec
ARIMA(0,0,0)(0,0,0)[12] intercept : AIC=1018.583, Time=0.03 sec
ARIMA(1,0,0)(0,0,0)[12] intercept : AIC=912.770, Time=0.08 sec
ARIMA(1,0,0)(0,0,1)[12] intercept : AIC=913.766, Time=0.18 sec
ARIMA(1,0,0)(1,0,1)[12] intercept : AIC=914.827, Time=0.83 sec
ARIMA(2,0,0)(0,0,0)[12] intercept : AIC=909.351, Time=0.26 sec
ARIMA(2,0,0)(1,0,0)[12] intercept : AIC=911.215, Time=0.47 sec
ARIMA(2,0,0)(0,0,1)[12] intercept : AIC=911.228, Time=0.15 sec
ARIMA(2,0,0)(1,0,1)[12] intercept : AIC=913.198, Time=0.89 sec
ARIMA(3,0,0)(0,0,0)[12] intercept : AIC=908.653, Time=0.12 sec
ARIMA(3,0,0)(1,0,0)[12] intercept : AIC=908.182, Time=1.63 sec
ARIMA(3,0,0)(2,0,0)[12] intercept : AIC=907.960, Time=1.76 sec
ARIMA(3,0,0)(2,0,1)[12] intercept : AIC=909.720, Time=0.74 sec
ARIMA(3,0,0)(1,0,1)[12] intercept : AIC=909.670, Time=0.23 sec
ARIMA(2,0,0)(2,0,0)[12] intercept : AIC=912.080, Time=0.50 sec
ARIMA(3,0,1)(2,0,0)[12] intercept : AIC=907.613, Time=1.37 sec
ARIMA(3,0,1)(1,0,0)[12] intercept : AIC=902.333, Time=0.97 sec
ARIMA(3,0,1)(0,0,0)[12] intercept : AIC=904.338, Time=1.29 sec
ARIMA(3,0,1)(1,0,1)[12] intercept : AIC=903.435, Time=1.64 sec
ARIMA(3,0,1)(0,0,1)[12] intercept : AIC=902.970, Time=0.40 sec
ARIMA(3,0,1)(2,0,1)[12] intercept : AIC=905.269, Time=1.29 sec
ARIMA(2,0,1)(1,0,0)[12] intercept : AIC=915.649, Time=0.24 sec
ARIMA(3,0,2)(1,0,0)[12] intercept : AIC=inf, Time=0.54 sec
ARIMA(2,0,2)(1,0,0)[12] intercept : AIC=inf, Time=0.44 sec
ARIMA(3,0,1)(1,0,0)[12] intercept : AIC=922.666, Time=0.30 sec

```

```

Best model: ARIMA(3,0,1)(1,0,0)[12] intercept
Total fit time: 18.306 seconds

```

```
Optimal SARIMA model parameters:
```

SARIMAX Results

```

=====
Dep. Variable:          y      No. Observations:
Model:                SARIMAX(3, 0, 1)x(1, 0, [], 12)  Log Likelihood
Date:                  Fri, 06 Feb 2026      AIC

```

```

Time:                                08:17:49    BIC
Sample:                             05-01-2017    HQIC
                                     - 09-01-2019
Covariance Type:                    opg
=====
              coef      std err          z      P>|z|      [0.025      0.975]
-----
intercept    3.814e+06    2.46e-07    1.55e+13    0.000    3.81e+06    3.81e+06
ar.L1         1.3464      0.320      4.206    0.000      0.719      1.974
ar.L2        -0.3081      0.588     -0.524    0.601     -1.461      0.845
ar.L3        -0.3604      0.324     -1.114    0.265     -0.995      0.274
ma.L1        -0.9519      0.473     -2.012    0.044     -1.879     -0.025
ar.S.L12     -0.2877      0.304     -0.947    0.344     -0.883      0.308
sigma2       1.55e+12    2.96e-13    5.24e+24    0.000    1.55e+12    1.55e+12
=====
Ljung-Box (L1) (Q):                0.03    Jarque-Bera (JB):

```

```

from statsmodels.tsa.statespace.sarimax import SARIMAX

# 4. Train the SARIMA Model
print("Retraining the SARIMA model with updated training data and parameter

# Retrieve the optimal parameters found by the most recent auto_arima on tr
# optimal_order = (p, d, q)
# optimal_seasonal_order = (P, D, Q, S)

# Instantiate the SARIMAX model
# The 'enforce_stationarity=False' and 'enforce_invertibility=False' arguments
# used to allow more flexibility in model fitting, especially when auto_arima
# has already determined differencing terms 'd' and 'D'.

model = SARIMAX(train_data,
                order=optimal_order,
                seasonal_order=optimal_seasonal_order,
                enforce_stationarity=False,
                enforce_invertibility=False)

# Fit the model to the training data
# suppress_warnings is used to clean up the output.
model_fit = model.fit(dispatch=False, low_memory=True)

print("SARIMA model retraining completed.")
print(model_fit.summary())

```

Retraining the SARIMA model with updated training data and parameters...
SARIMA model retraining completed.

SARIMAX Results

```

=====
Dep. Variable:      Electricity_consumption    No. Observations:
Model:              SARIMAX(3, 0, 1)x(1, 0, [], 12)    Log Likelihood
Date:              Fri, 06 Feb 2026    AIC
Time:              08:18:02    BIC
Sample:            05-01-2017    HQIC

```

- 09-01-2019

Covariance Type:

approx

	coef	std err	z	P> z	[0.025	0.975]
ar.L1	0.3963	0.377	1.051	0.293	-0.343	1.135
ar.L2	-0.4065	0.323	-1.259	0.208	-1.039	0.226
ar.L3	0.3438	0.199	1.730	0.084	-0.046	0.733
ma.L1	-0.4714	0.308	-1.532	0.126	-1.075	0.132
ar.S.L12	0.9576	0.033	29.174	0.000	0.893	1.022
sigma2	2.121e+12	7.15e-18	2.97e+29	0.000	2.12e+12	2.12e+12
=====						
Ljung-Box (L1) (Q):			0.02	Jarque-Bera (JB):		1
Prob(Q):			0.88	Prob(JB):		0
Heteroskedasticity (H):			0.51	Skew:		-0
Prob(H) (two-sided):			0.47	Kurtosis:		2
=====						

Warnings:

- [1] Covariance matrix calculated using numerical (complex-step) differentiation.
 [2] Covariance matrix is singular or near-singular, with condition number 1.11e+

```
import pandas as pd
```

```
# 1. Extract the 'Temperature' column from df_energy aligned with train_data
# First, ensure df_energy is indexed by Date for easy alignment
# (This should already be the case from previous preprocessing steps)
```

```
# Filter df_energy to match the index of train_data
exog_train = df_energy.loc[train_data.index, 'Temperature']
```

```
print("Exogenous variable 'exog_train' created.")
print("Shape of exog_train:", exog_train.shape)
print("First 5 rows of exog_train:")
print(exog_train.head())
```

```
Exogenous variable 'exog_train' created.
Shape of exog_train: (29,)
First 5 rows of exog_train:
2017-05-01    33.7
2017-06-01    36.6
2017-07-01    39.2
2017-08-01    38.2
2017-09-01    35.5
Freq: MS, Name: Temperature, dtype: float64
```

```
import pmdarima as pm
```

```
# 2. Determine SARIMAX Parameters (p, d, q, P, D, Q, S) using auto_arima w:
print("Determining optimal SARIMAX parameters with exogenous variable 'Tem
```

```
# Use train_data for auto_arima and exog_train as the exogenous variable.
# Ensure m=12 for monthly seasonality
```

```

# Ensure m=12 for monthly seasonality.

# Given the short length of the training series, setting max_p, max_q, max_

# Ensure the series has a DatetimeIndex for auto_arima
if not isinstance(train_data.index, pd.DatetimeIndex):
    train_data.index = pd.to_datetime(train_data.index)

# Fit auto_arima model with exogenous variable
# Max order for p, q, P, Q are set conservatively due to short series length
# D (seasonal differencing) is explicitly set to 0.
# d (non-seasonal differencing) is left for auto_arima to determine.

optimal_model_sarimax = pm.auto_arima(train_data,
                                     exog=exog_train,
                                     start_p=0, start_q=0,
                                     test='adf',          # use adftest to find best
                                     max_p=3, max_q=3,    # maximum p and q
                                     m=12,                # frequency of series
                                     start_P=0, start_Q=0,
                                     max_P=2, max_Q=2,    # maximum P and Q
                                     D=0,                 # Explicitly set D=0
                                     d=None,              # Let auto_arima decide
                                     trace=True,
                                     error_action='ignore',
                                     suppress_warnings=True,
                                     stepwise=True,
                                     n_jobs=-1)          # use all available cores

print("\nOptimal SARIMAX model parameters with exogenous variable:")
print(optimal_model_sarimax.summary())

# 3. Store the identified optimal non-seasonal and seasonal orders
optimal_order_sarimax = optimal_model_sarimax.order
optimal_seasonal_order_sarimax = optimal_model_sarimax.seasonal_order

print(f"\nNon-seasonal order (p,d,q): {optimal_order_sarimax}")
print(f"Seasonal order (P,D,Q,S): {optimal_seasonal_order_sarimax}")

```

```

Determining optimal SARIMAX parameters with exogenous variable 'Temperature' using
Performing stepwise search to minimize aic
ARIMA(0,0,0)(0,0,0)[12] intercept : AIC=940.399, Time=0.02 sec
/usr/local/lib/python3.12/dist-packages/pmdarima/arma/_validation.py:76: UserWarning:
warnings.warn('stepwise model cannot be fit in parallel (n_jobs=%i). '
ARIMA(1,0,0)(1,0,0)[12] intercept : AIC=912.857, Time=0.24 sec
ARIMA(0,0,1)(0,0,1)[12] intercept : AIC=925.454, Time=0.15 sec
ARIMA(0,0,0)(0,0,0)[12] intercept : AIC=1018.583, Time=0.03 sec
ARIMA(1,0,0)(0,0,0)[12] intercept : AIC=912.770, Time=0.05 sec
ARIMA(1,0,0)(0,0,1)[12] intercept : AIC=913.766, Time=0.09 sec
ARIMA(1,0,0)(1,0,1)[12] intercept : AIC=914.827, Time=0.11 sec
ARIMA(2,0,0)(0,0,0)[12] intercept : AIC=909.351, Time=0.03 sec
ARIMA(2,0,0)(1,0,0)[12] intercept : AIC=911.215, Time=0.25 sec
ARIMA(2,0,0)(0,0,1)[12] intercept : AIC=911.228, Time=0.12 sec

```

```

ARIMA(2,0,0)(1,0,1)[12] intercept : AIC=913.198, Time=0.23 sec
ARIMA(3,0,0)(0,0,0)[12] intercept : AIC=908.653, Time=0.05 sec
ARIMA(3,0,0)(1,0,0)[12] intercept : AIC=908.182, Time=0.58 sec
ARIMA(3,0,0)(2,0,0)[12] intercept : AIC=907.960, Time=2.24 sec
ARIMA(3,0,0)(2,0,1)[12] intercept : AIC=909.720, Time=1.76 sec
ARIMA(3,0,0)(1,0,1)[12] intercept : AIC=909.670, Time=0.56 sec
ARIMA(2,0,0)(2,0,0)[12] intercept : AIC=912.080, Time=1.15 sec
ARIMA(3,0,1)(2,0,0)[12] intercept : AIC=907.613, Time=4.53 sec
ARIMA(3,0,1)(1,0,0)[12] intercept : AIC=902.333, Time=0.64 sec
ARIMA(3,0,1)(0,0,0)[12] intercept : AIC=904.338, Time=0.16 sec
ARIMA(3,0,1)(1,0,1)[12] intercept : AIC=903.435, Time=0.64 sec
ARIMA(3,0,1)(0,0,1)[12] intercept : AIC=902.970, Time=0.33 sec
ARIMA(3,0,1)(2,0,1)[12] intercept : AIC=905.269, Time=1.34 sec
ARIMA(2,0,1)(1,0,0)[12] intercept : AIC=915.649, Time=0.25 sec
ARIMA(3,0,2)(1,0,0)[12] intercept : AIC=inf, Time=0.51 sec
ARIMA(2,0,2)(1,0,0)[12] intercept : AIC=inf, Time=0.48 sec
ARIMA(3,0,1)(1,0,0)[12] : AIC=922.666, Time=0.25 sec

```

Best model: ARIMA(3,0,1)(1,0,0)[12] intercept

Total fit time: 16.838 seconds

Optimal SARIMAX model parameters with exogenous variable:

SARIMAX Results

```

=====
Dep. Variable:          y      No. Observations:
Model:                SARIMAX(3, 0, 1)x(1, 0, [], 12)  Log Likelihood
Date:                  Fri, 06 Feb 2026              AIC
Time:                   08:19:04                     BIC
Sample:                05-01-2017                   HQIC
                    - 09-01-2019

Covariance Type:                opg
=====

```

	coef	std err	z	P> z	[0.025	0.975]
intercept	3.814e+06	2.46e-07	1.55e+13	0.000	3.81e+06	3.81e+06
ar.L1	1.3464	0.320	4.206	0.000	0.719	1.974
ar.L2	-0.3081	0.588	-0.524	0.601	-1.461	0.845
ar.L3	-0.3604	0.324	-1.114	0.265	-0.995	0.274
ma.L1	-0.9519	0.473	-2.012	0.044	-1.879	-0.025
ar.S.L12	-0.2877	0.304	-0.947	0.344	-0.883	0.308
sigma2	1.55e+12	2.96e-13	5.24e+24	0.000	1.55e+12	1.55e+12

```

=====
Liung-Box (L1) (0):                0.03   Jarque-Bera (JB):

```

```
from statsmodels.tsa.statespace.sarimax import SARIMAX
```

```
# 4. Instantiate a SARIMAX model
```

```
print("Instantiating SARIMAX model with optimal parameters and exogenous v
```

```

model_sarimax = SARIMAX(train_data,
                        order=optimal_order_sarimax,
                        seasonal_order=optimal_seasonal_order_sarimax,
                        exog=exog_train,
                        enforce_stationarity=False,
                        enforce_invertibility=False)

```

```
# 5. Fit the SARIMAX model to the training data
print("Fitting SARIMAX model...")
model_sarimax_fit = model_sarimax.fit(dispatch=False, low_memory=True)

print("SARIMAX model training completed.")
print(model_sarimax_fit.summary())
```

Instantiating SARIMAX model with optimal parameters and exogenous variable...
 Fitting SARIMAX model...
 SARIMAX model training completed.

SARIMAX Results

```
=====
Dep. Variable:          Electricity_consumption    No. Observations:
Model:                SARIMAX(3, 0, 1)x(1, 0, [], 12)    Log Likelihood
Date:                  Fri, 06 Feb 2026              AIC
Time:                  08:19:16                      BIC
Sample:                05-01-2017                   HQIC
                    - 09-01-2019
Covariance Type:        approx
=====
```

	coef	std err	z	P> z	[0.025	0.975]
Temperature	3.098e+05	nan	nan	nan	nan	nan
ar.L1	0.0538	nan	nan	nan	nan	nan
ar.L2	0.4726	nan	nan	nan	nan	nan
ar.L3	-0.2966	nan	nan	nan	nan	nan
ma.L1	1.1253	nan	nan	nan	nan	nan
ar.S.L12	-0.6892	0.276	-2.501	0.012	-1.229	-0.149
sigma2	7.152e+11	nan	nan	nan	nan	nan

```
=====
Ljung-Box (L1) (Q):                0.51    Jarque-Bera (JB):                0
Prob(Q):                          0.48    Prob(JB):                          0
Heteroskedasticity (H):            0.50    Skew:                              -0
Prob(H) (two-sided):              0.47    Kurtosis:                          2
=====
```

Warnings:

[1] Covariance matrix calculated using numerical (complex-step) differentiation.
 [2] Covariance matrix is singular or near-singular, with condition number 1e+29

```
import matplotlib.pyplot as plt
import seaborn as sns
from statsmodels.graphics.tsaplots import plot_acf
from scipy import stats
import numpy as np
```

```
# 1. Manually generate diagnostics plots for the model_fit (SARIMA)
print("Manually generating diagnostics plots for SARIMA model...")
```

```
residuals = model_fit.resid
```

```
fig, axes = plt.subplots(4, 1, figsize=(12, 10))
```

```
fig.suptitle('SARIMA Model Residual Diagnostics', y=1.02)

# Standardized Residuals Plot
axes[0].plot(residuals)
axes[0].set_title('Standardized Residuals')
axes[0].grid(True)

# Histogram plus KDE
sns.histplot(residuals, kde=True, ax=axes[1])
axes[1].set_title('Histogram and KDE of Residuals')

# Normal QQ-plot
stats.probplot(residuals, dist="norm", plot=axes[2])
axes[2].set_title('Normal Q-Q plot of Residuals')

# Correlogram (ACF plot)
# Set 'lags' explicitly to a value appropriate for the training data length
# The residuals length is len(train_data) = 29. A common rule is lags < N/2
# Using lags=12 as it's the seasonal period and less than 29/2 (14.5).
# Also ensure there are enough points for the ACF plot calculation.
if len(residuals) > 12: # Ensure enough data points for lags=12
    plot_acf(residuals, lags=12, ax=axes[3])
    axes[3].set_title('Correlogram (ACF) of Residuals')
else:
    axes[3].text(0.5, 0.5, 'Not enough data for ACF plot with 12 lags', horizontalalignment='center')
    axes[3].set_title('Correlogram (ACF) of Residuals')

plt.tight_layout(rect=[0, 0.03, 1, 0.98])
plt.show()
print("SARIMA model diagnostics plots displayed.")
```

Manually generating diagnostics plots for SARIMA model...

SARIMA Model Residual Diagnostics



```
import matplotlib.pyplot as plt
import seaborn as sns
from statsmodels.graphics.tsaplots import plot_acf
from scipy import stats
import numpy as np

# 2. Manually generate diagnostics plots for the model_sarimax_fit (SARIMA)
print("Manually generating diagnostics plots for SARIMAX model...")

residuals_sarimax = model_sarimax_fit.resid

fig, axes = plt.subplots(4, 1, figsize=(12, 10))
fig.suptitle('SARIMAX Model Residual Diagnostics', y=1.02)

# Standardized Residuals Plot
axes[0].plot(residuals_sarimax)
axes[0].set_title('Standardized Residuals')
axes[0].grid(True)

# Histogram plus KDE
sns.histplot(residuals_sarimax, kde=True, ax=axes[1])
axes[1].set_title('Histogram and KDE of Residuals')

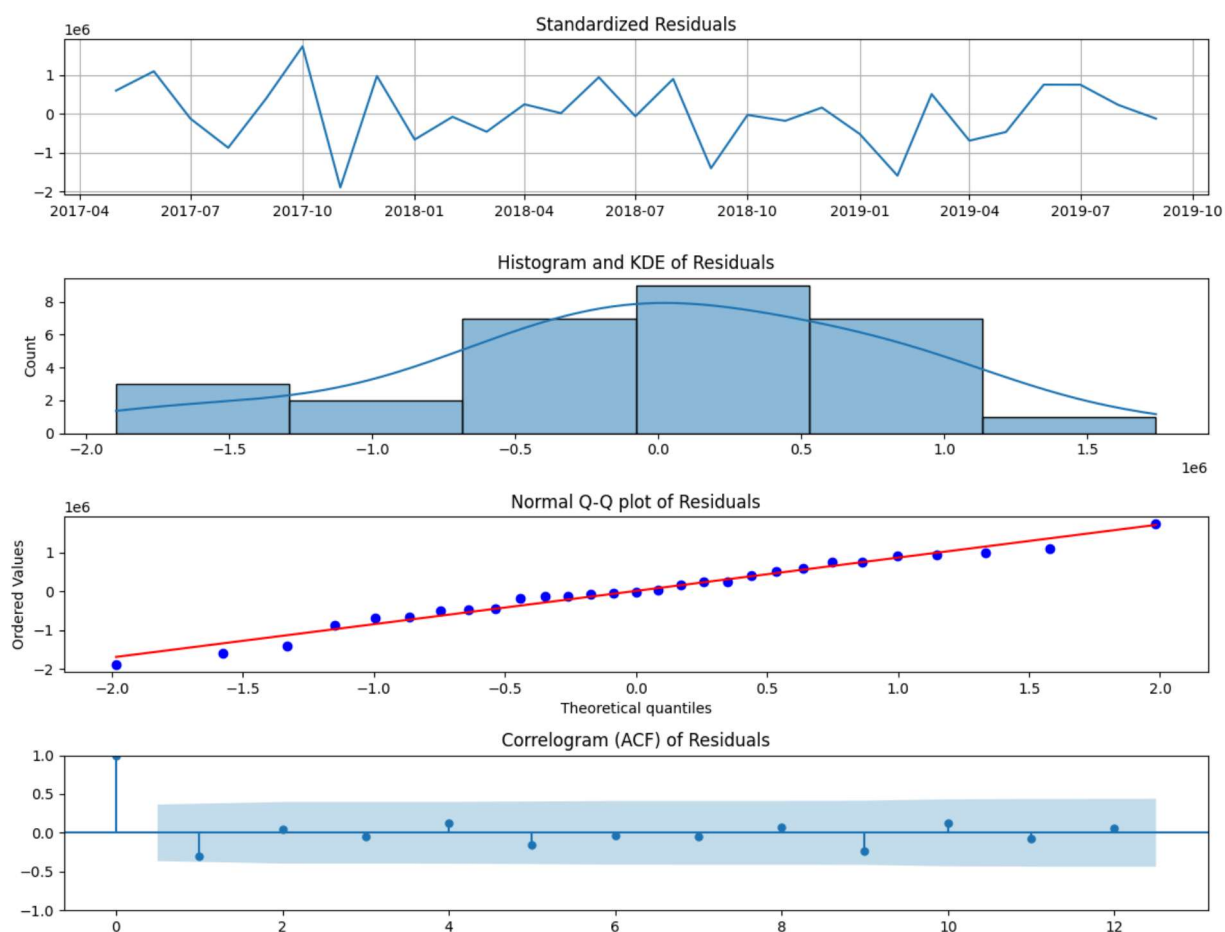
# Normal QQ-plot
stats.probplot(residuals_sarimax, dist="norm", plot=axes[2])
axes[2].set_title('Normal Q-Q plot of Residuals')

# Correlogram (ACF plot)
# Set 'lags' explicitly to a value appropriate for the training data length
# The residuals length is len(train_data) = 29. A common rule is lags < N/2
# Using lags=12 as it's the seasonal period and less than 29/2 (14.5).
# Also ensure there are enough points for the ACF plot calculation.
if len(residuals_sarimax) > 12: # Ensure enough data points for lags=12
    plot_acf(residuals_sarimax, lags=12, ax=axes[3])
    axes[3].set_title('Correlogram (ACF) of Residuals')
else:
    axes[3].text(0.5, 0.5, 'Not enough data for ACF plot with 12 lags', horizontalalignment='center')
    axes[3].set_title('Correlogram (ACF) of Residuals')

plt.tight_layout(rect=[0, 0.03, 1, 0.98])
plt.show()
print("SARIMAX model diagnostics plots displayed.")
```

Manually generating diagnostics plots for SARIMAX model...

SARIMAX Model Residual Diagnostics



SARIMAX model diagnostics plots displayed.

```
import pandas as pd
```

```
# 3. Get the predictions for the test set for the model_fit (SARIMA)
print("Generating SARIMA model predictions for the test set...")
```

```
sarima_predictions = model_fit.predict(start=test_data.index[0], end=test_data.index[-1])
```

```
print("SARIMA predictions generated.")
print("First 5 SARIMA predictions:")
print(sarima_predictions.head())
print("Last 5 SARIMA predictions:")
print(sarima_predictions.tail())
```

Generating SARIMA model predictions for the test set...

SARIMA predictions generated.

First 5 SARIMA predictions:

2019-10-01 7.391576e+06

2019-11-01 7.203287e+06

2019-12-01 6.349887e+06

Freq: MS, Name: predicted_mean, dtype: float64

Last 5 SARIMA predictions:

```

2019-10-01    7.391576e+06
2019-11-01    7.203287e+06
2019-12-01    6.349887e+06
Freq: MS, Name: predicted_mean, dtype: float64

```

```

import pandas as pd

# 4. Get the predictions for the test set for the model_sarimax_fit (SARIMAX)
print("Generating SARIMAX model predictions for the test set...")

# Ensure df_energy is indexed by Date for easy alignment (already done in previous cell)

# Extract the 'Temperature' column from df_energy aligned with test_data for exogenous variable
exog_test = df_energy.loc[test_data.index, 'Temperature']

# Generate SARIMAX predictions, including the exogenous variable for the forecast
sarimax_predictions = model_sarimax_fit.predict(start=test_data.index[0],
                                                end=test_data.index[-1],
                                                exog=exog_test)

print("SARIMAX predictions generated.")
print("First 5 SARIMAX predictions:")
print(sarimax_predictions.head())
print("Last 5 SARIMAX predictions:")
print(sarimax_predictions.tail())

```

```

Generating SARIMAX model predictions for the test set...
SARIMAX predictions generated.
First 5 SARIMAX predictions:
2019-10-01    1.204140e+07
2019-11-01    8.496901e+06
2019-12-01    7.516974e+06
Freq: MS, Name: predicted_mean, dtype: float64
Last 5 SARIMAX predictions:
2019-10-01    1.204140e+07
2019-11-01    8.496901e+06
2019-12-01    7.516974e+06
Freq: MS, Name: predicted_mean, dtype: float64

```

```

from sklearn.metrics import mean_squared_error, mean_absolute_error
import numpy as np

# 5. Calculate evaluation metrics (RMSE, MAE, MAPE) for both models
print("Calculating evaluation metrics...")

# Ensure predictions and test_data are aligned
sarima_predictions = sarima_predictions.reindex(test_data.index)
sarimax_predictions = sarimax_predictions.reindex(test_data.index)

# RMSE
rmse_sarima = np.sqrt(mean_squared_error(test_data, sarima_predictions))
rmse_sarimax = np.sqrt(mean_squared_error(test_data, sarimax_predictions))

```

```

# MAE
mae_sarima = mean_absolute_error(test_data, sarima_predictions)
mae_sarimax = mean_absolute_error(test_data, sarimax_predictions)

# MAPE
def calculate_mape(y_true, y_pred):
    # Handle potential division by zero by filtering out actual zero values
    # or setting a small epsilon. Here, we'll filter.
    y_true, y_pred = np.array(y_true), np.array(y_pred)
    non_zero_indices = y_true != 0
    return np.mean(np.abs((y_true[non_zero_indices] - y_pred[non_zero_indices]) / y_true[non_zero_indices]))

mape_sarima = calculate_mape(test_data, sarima_predictions)
mape_sarimax = calculate_mape(test_data, sarimax_predictions)

# 6. Print the calculated evaluation metrics for both models
print("\n--- SARIMA Model Evaluation ---")
print(f"RMSE: {rmse_sarima:.2f}")
print(f"MAE: {mae_sarima:.2f}")
print(f"MAPE: {mape_sarima:.2f}%")

print("\n--- SARIMAX Model Evaluation ---")
print(f"RMSE: {rmse_sarimax:.2f}")
print(f"MAE: {mae_sarimax:.2f}")
print(f"MAPE: {mape_sarimax:.2f}%")

```

Calculating evaluation metrics...

--- SARIMA Model Evaluation ---

RMSE: 2121415.36

MAE: 1871464.44

MAPE: 27.03%

--- SARIMAX Model Evaluation ---

RMSE: 2460159.00

MAE: 2312584.40

MAPE: 38.64%

```
import matplotlib.pyplot as plt
```

```
# 7. Plot the train_data, test_data, SARIMA predictions, and SARIMAX predictions
print("Generating plot of actual vs. predicted values...")
```

```

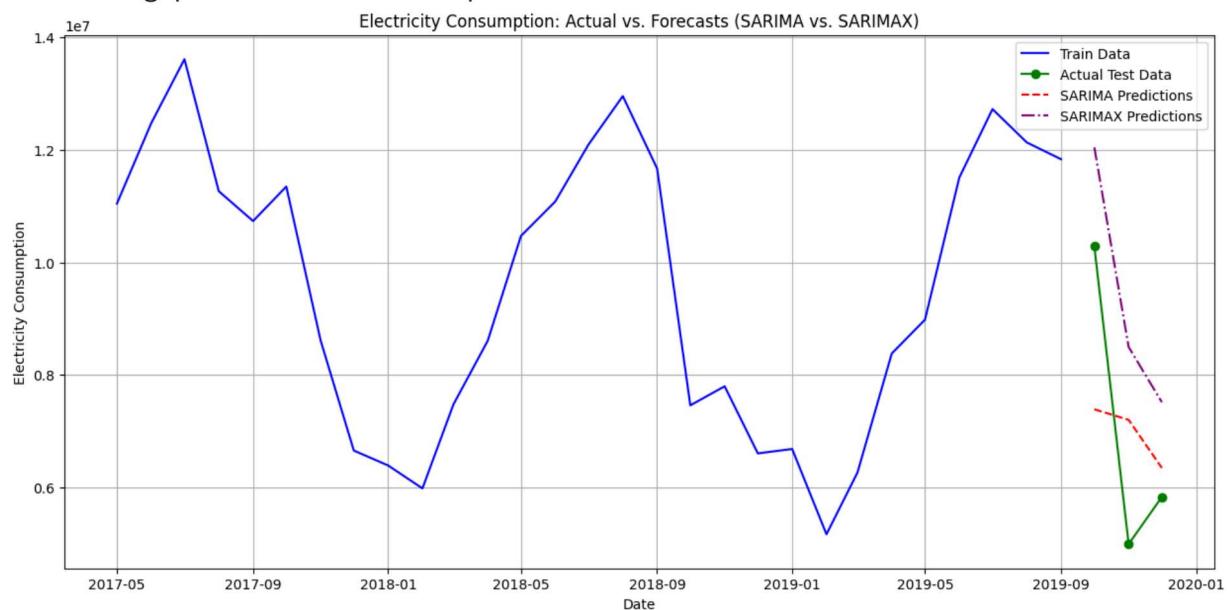
plt.figure(figsize=(15, 7))
plt.plot(train_data.index, train_data, label='Train Data', color='blue')
plt.plot(test_data.index, test_data, label='Actual Test Data', color='green')
plt.plot(sarima_predictions.index, sarima_predictions, label='SARIMA Predictions', color='red')
plt.plot(sarimax_predictions.index, sarimax_predictions, label='SARIMAX Predictions', color='purple')

plt.title('Electricity Consumption: Actual vs. Forecasts (SARIMA vs. SARIMAX)')
plt.xlabel('Date')

```

```
plt.ylabel('Electricity Consumption')
plt.legend()
plt.grid(True)
plt.show()
print("Plot of actual vs. predicted values displayed.")
```

Generating plot of actual vs. predicted values...



Plot of actual vs. predicted values displayed.